

Ancient Sheep Genomes reveal four Millennia of North European Short-Tailed Sheep in the Baltic Sea region

Supplementary Figures

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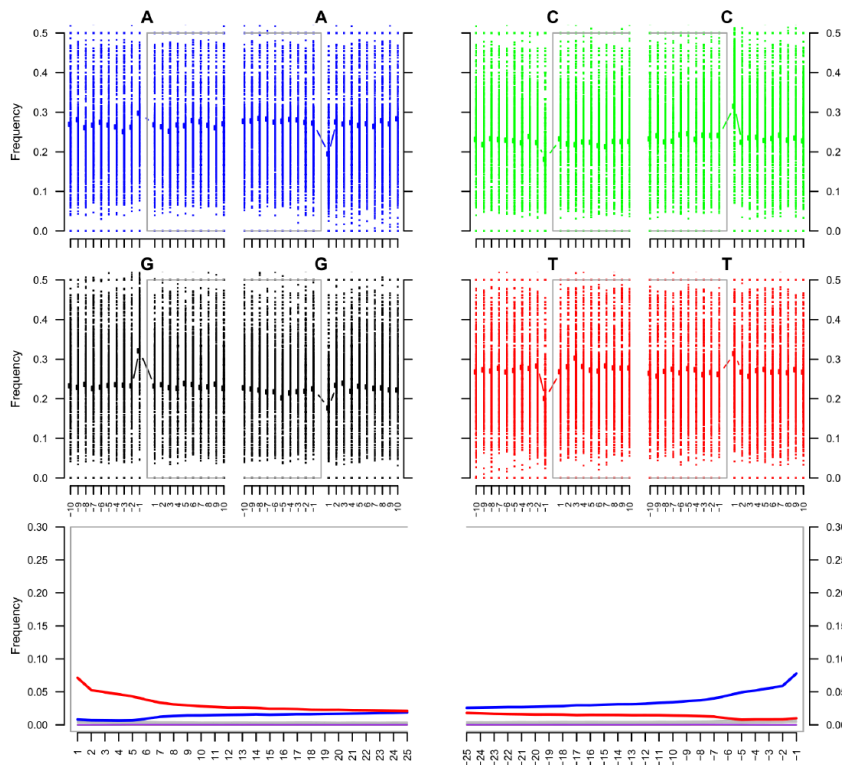
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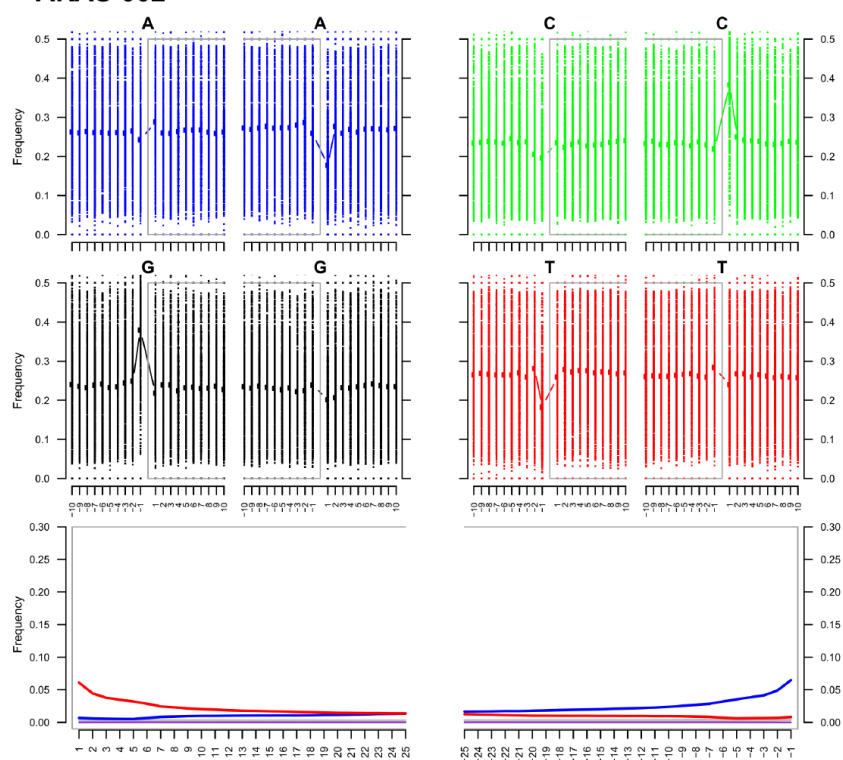
AKAS-001

out.extendedFrag.fastq.sorted



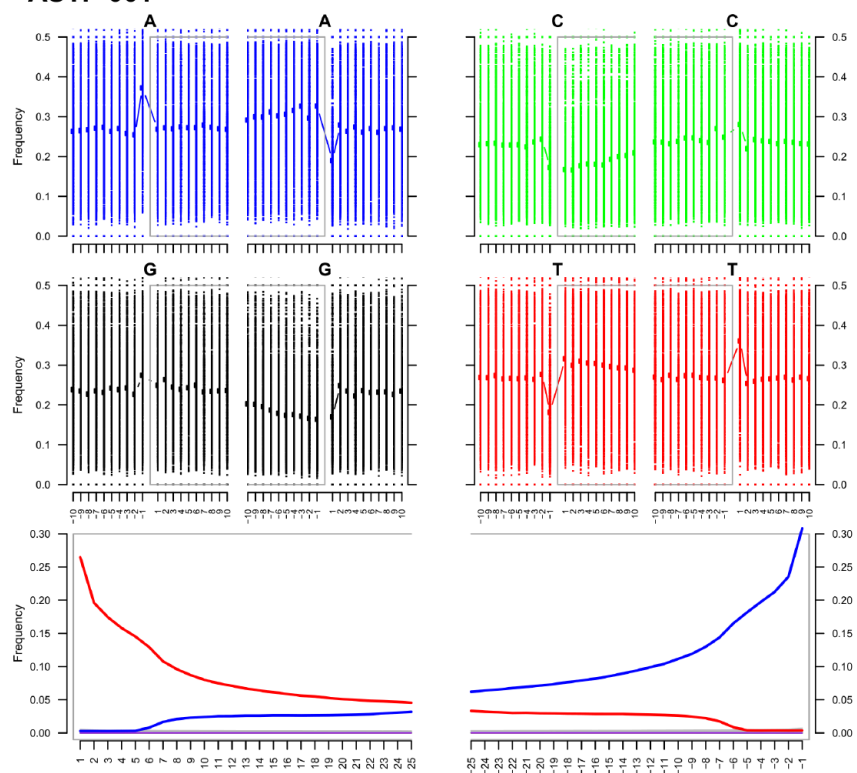
AKAS-002

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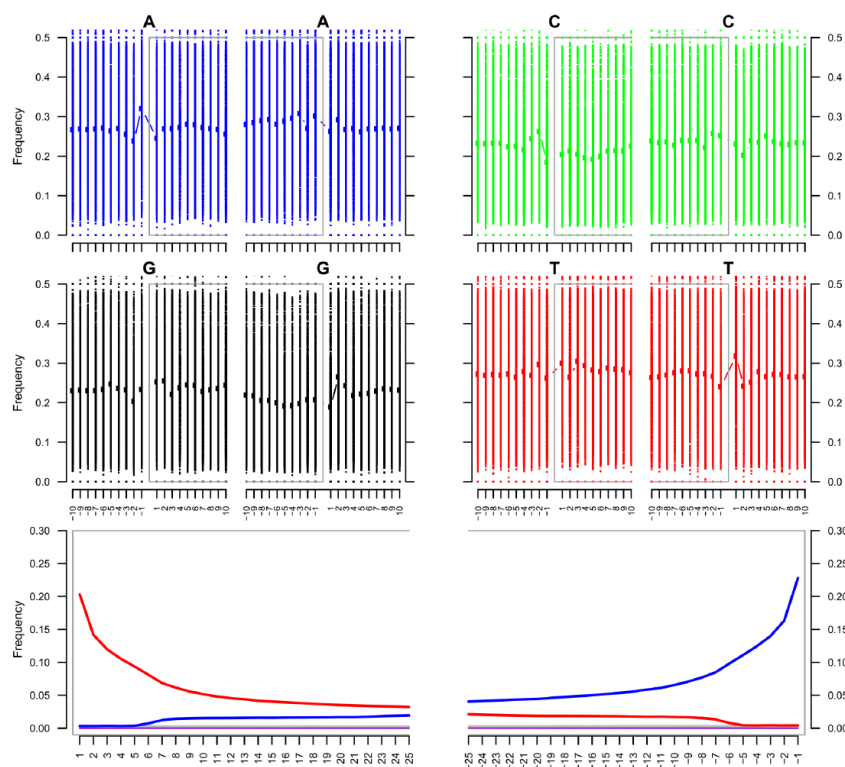
ASTF-001

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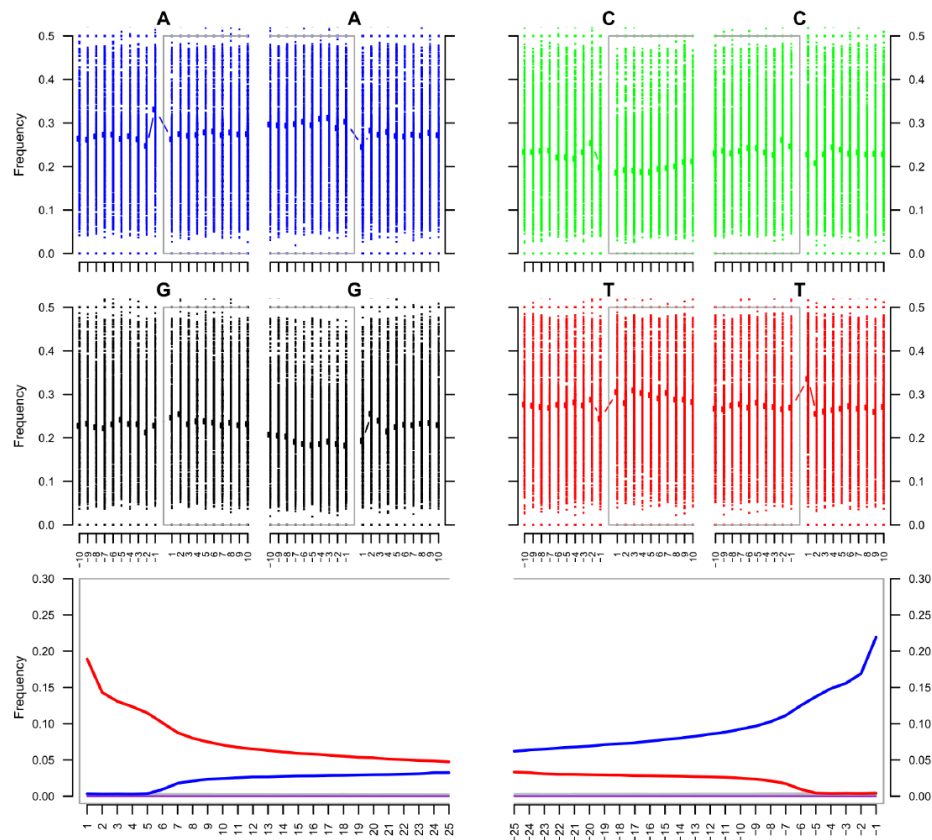


ASTF-002

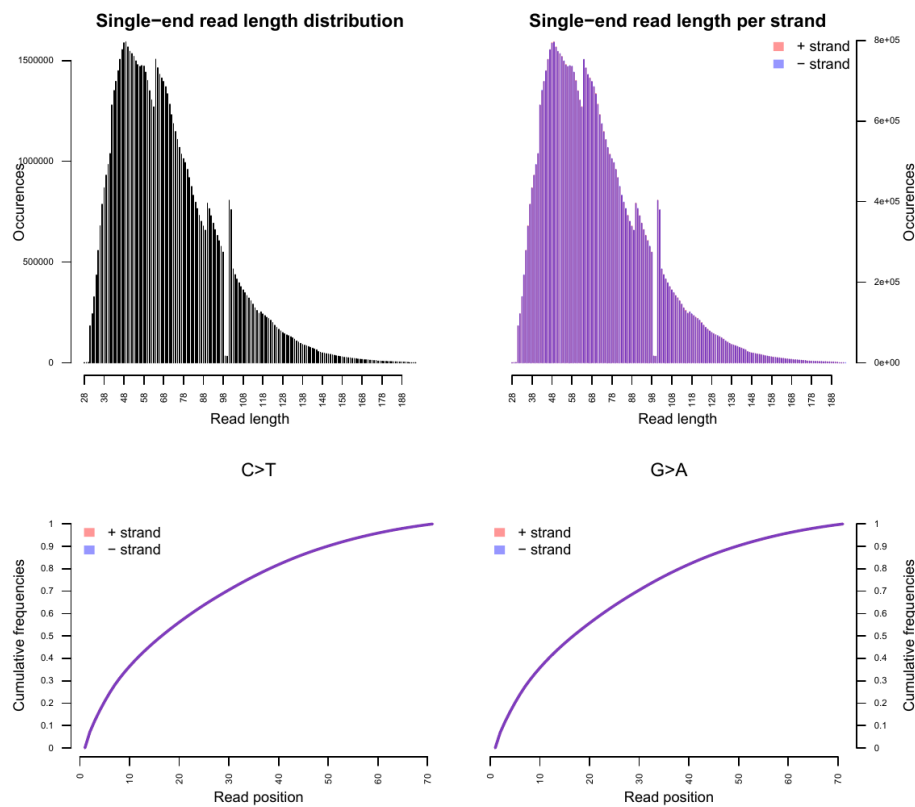
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**ASTF-003**

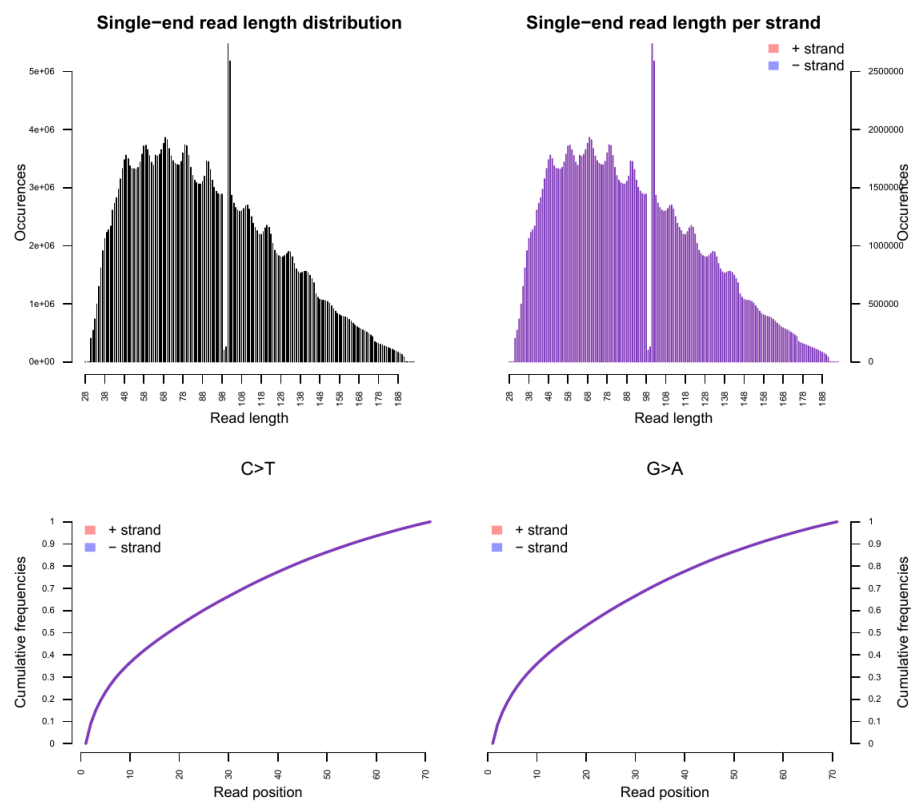
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**Supplementary Figure 1:** mapDamage misincorporation plots for the 5 ancient samples.

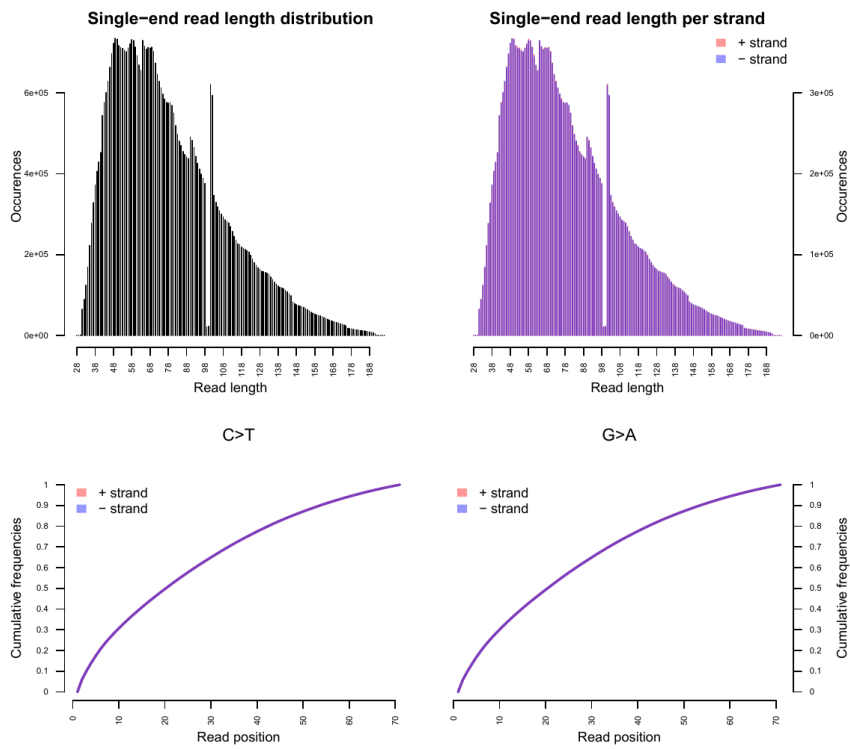
ASTF-001



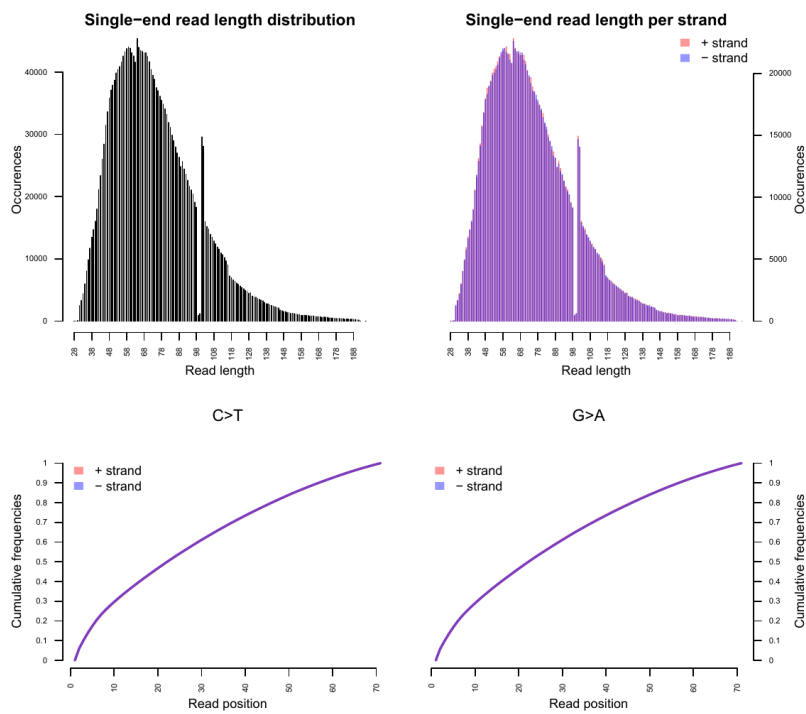
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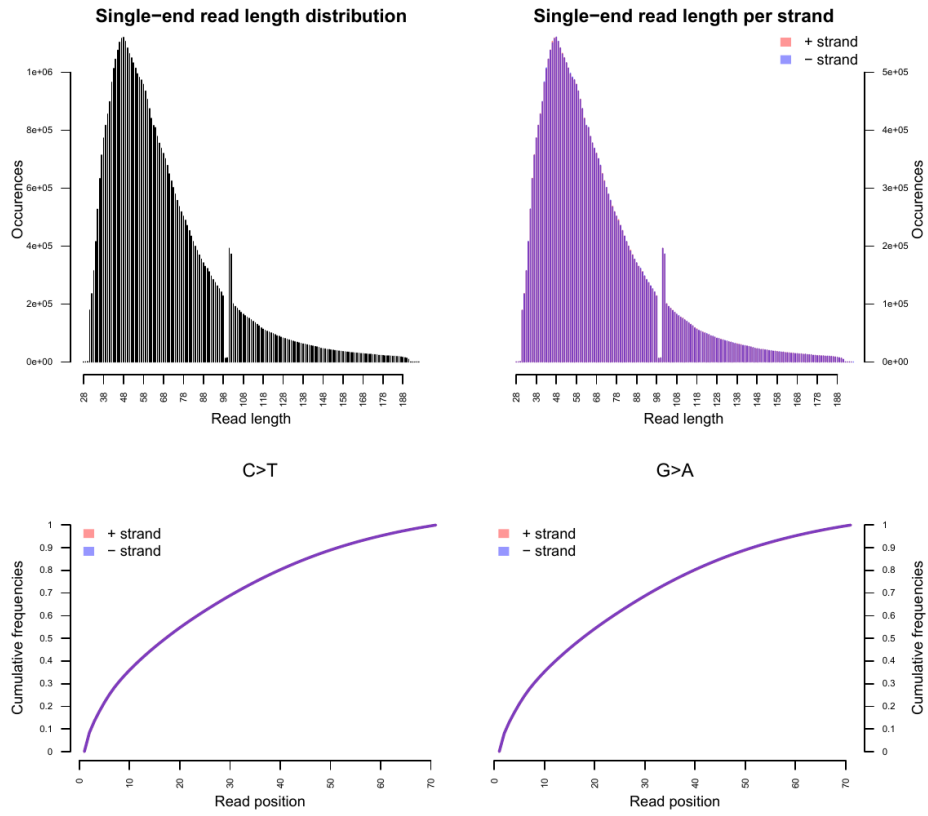
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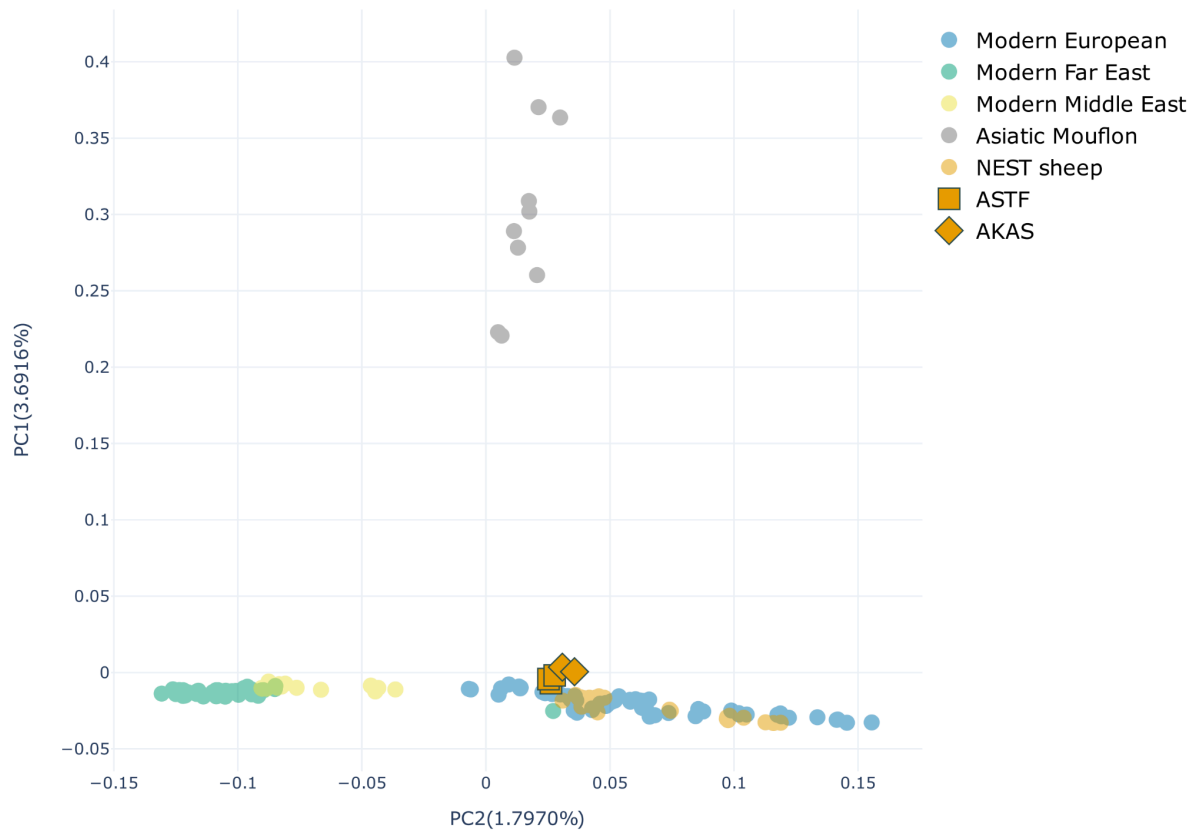
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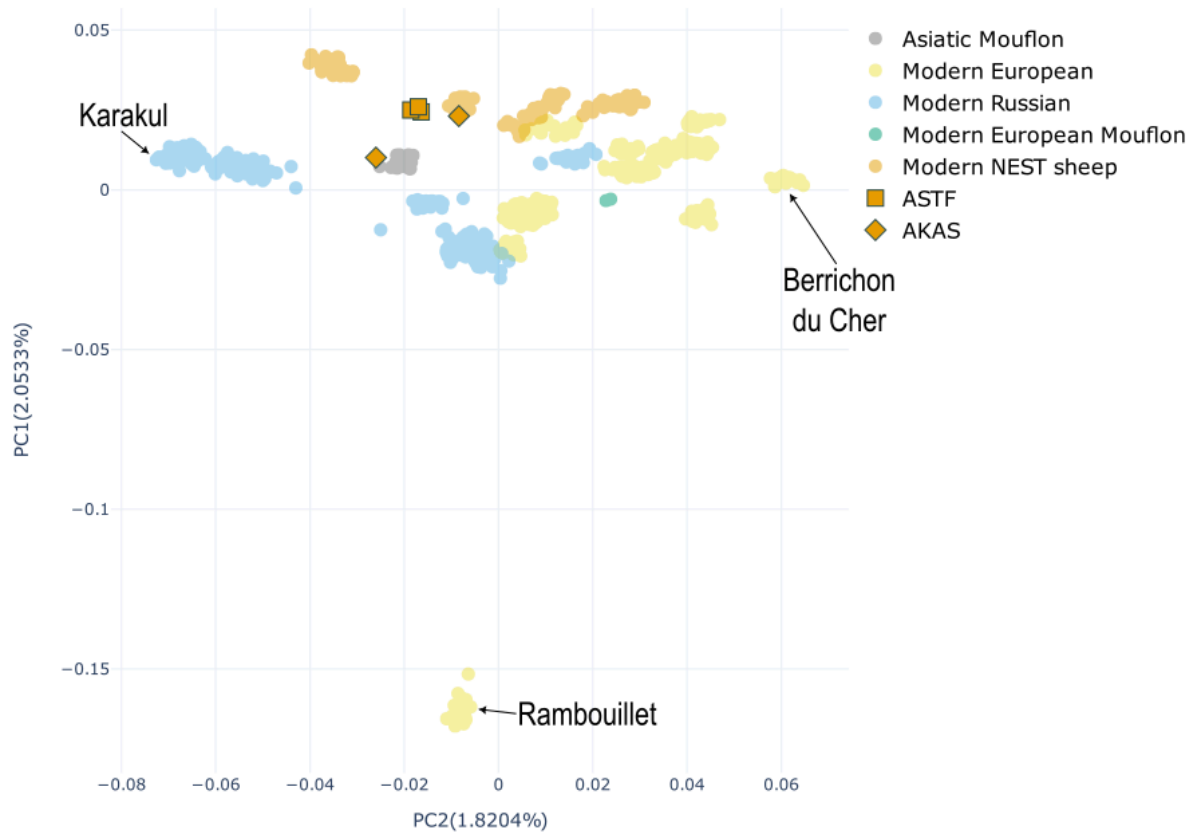
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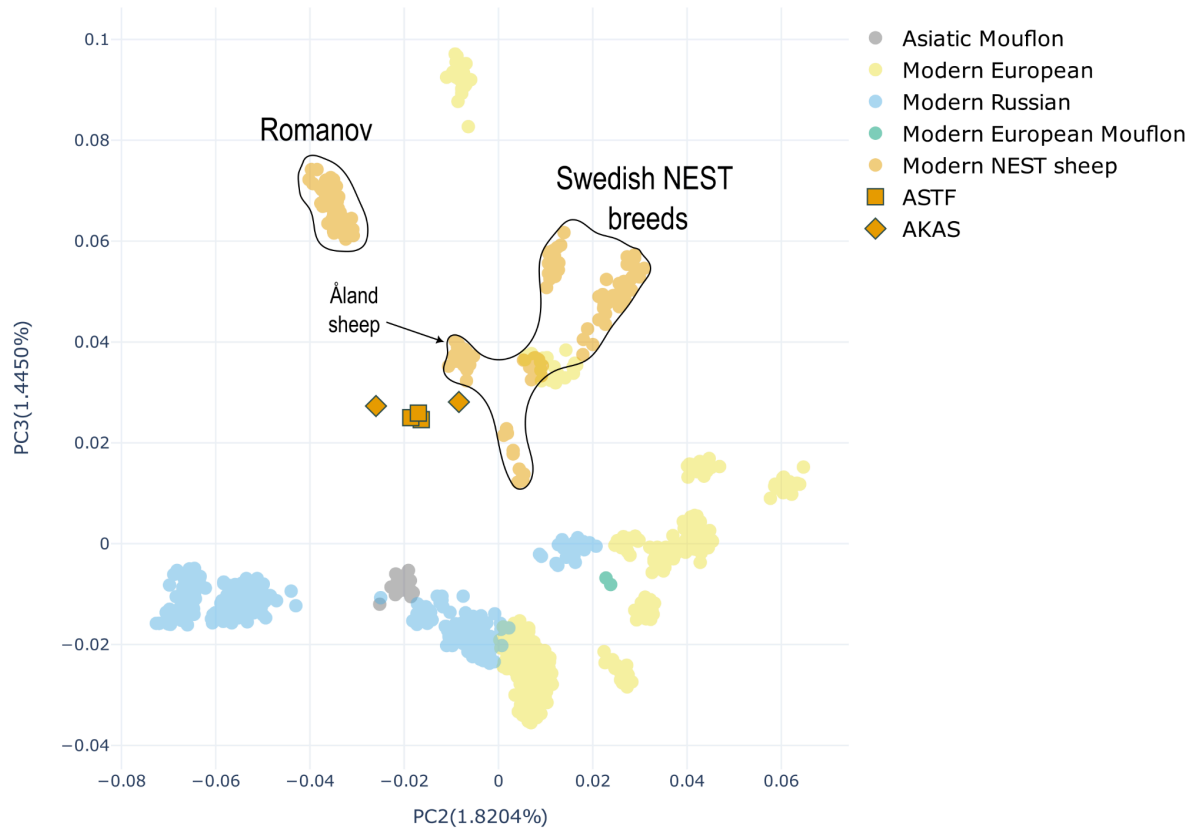
Supplementary Figure 2: mapDamage fragment length plots for the 5 ancient samples.



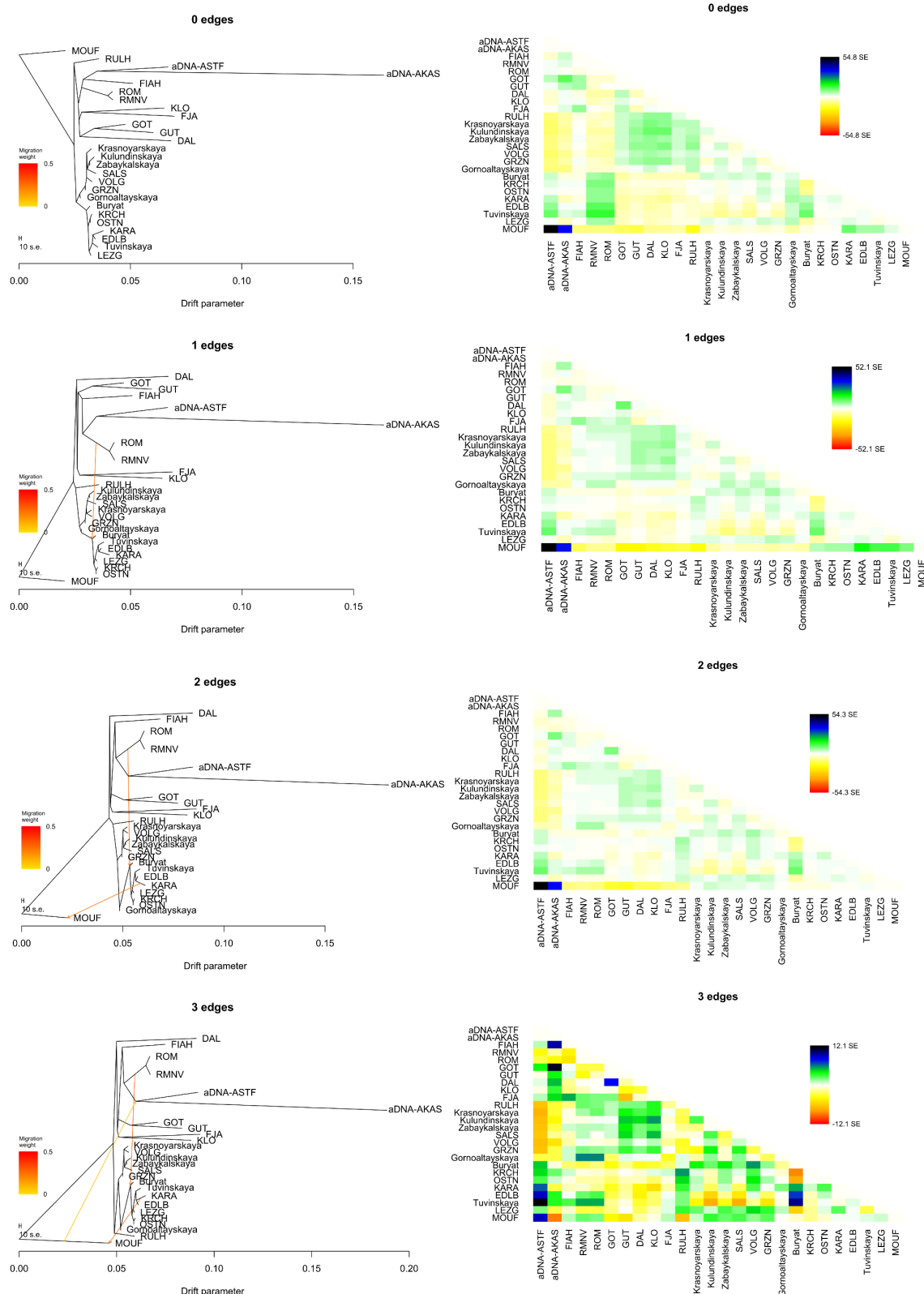
Supplementary Figure 3: WGS PCA showing PC 1 vs PC 2, percentage of explained variation is shown within parentheses of axis titles.



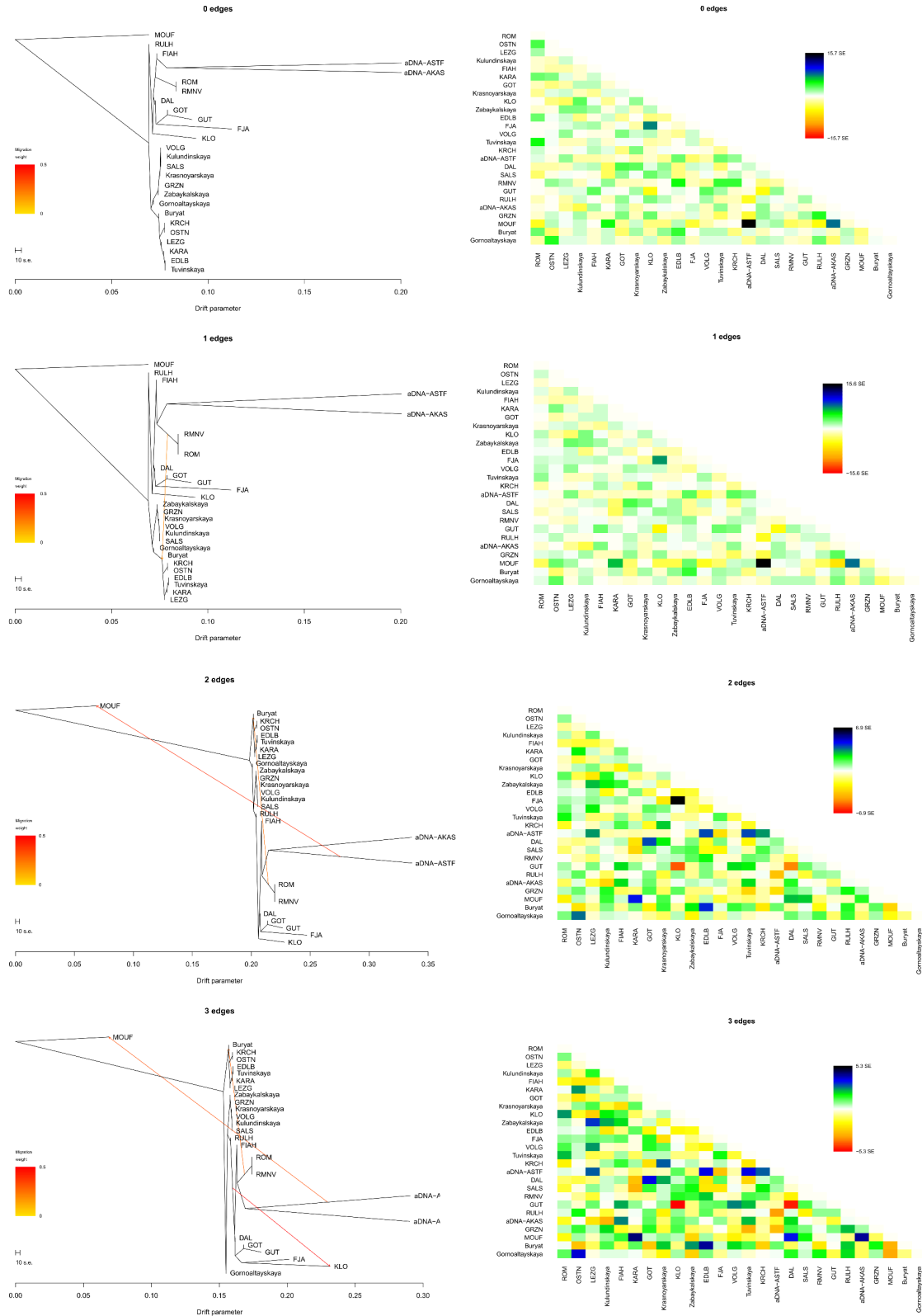
Supplementary Figure 4: SNPCHP PCA showing PC 1 vs PC 2, percentage of explained variation is shown within parentheses of axis titles.



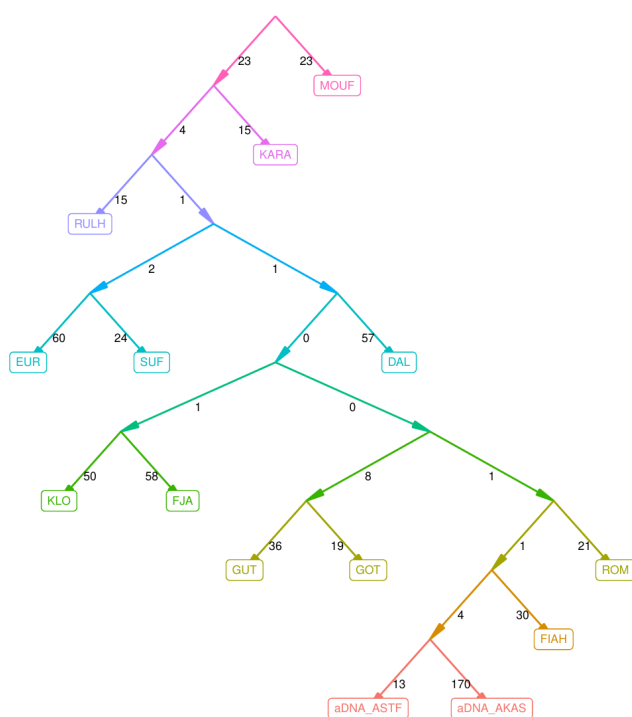
Supplementary Figure 5: SNPCHP PCA showing PC 2 vs PC 3, percentage of explained variation is shown within parentheses of axis titles. The Swedish NEST breeds, Romanov and the ancient samples are pointed out.



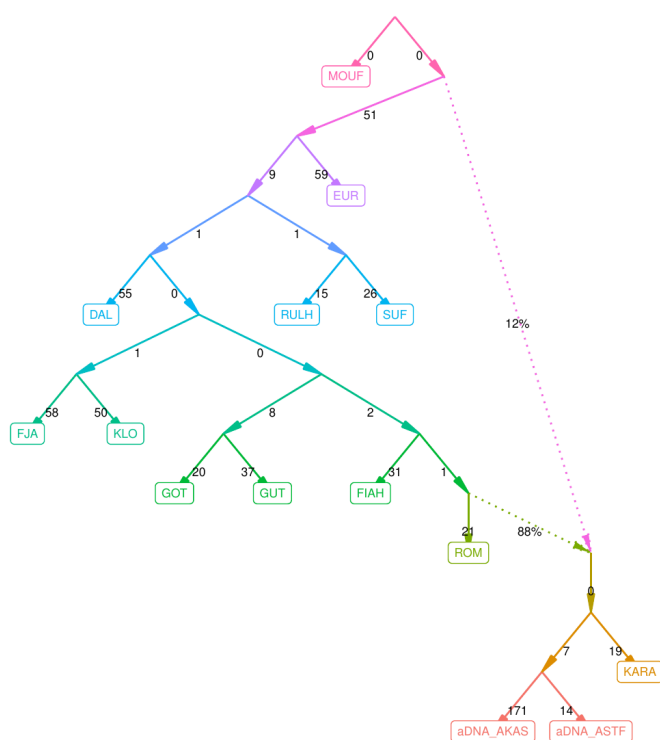
Supplementary Figure 8: OrientAGraph trees from the SNPCHP Panel, 0-3 migration edges, All Russian and Swedish breeds in SNPCHP panel, Asiatic mouflon as well as Ancients.



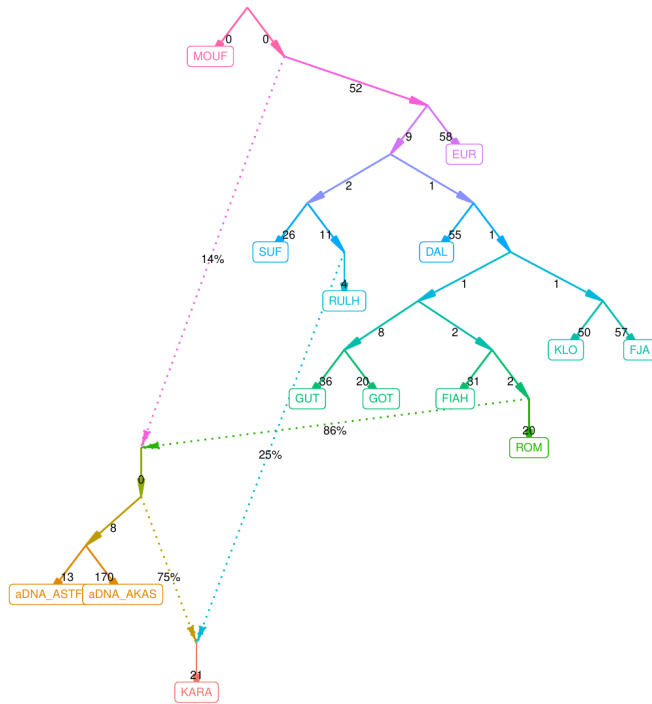
Supplementary Figure 9: OrientAGraph trees from the SNPCHP Panel using only one random individual from each modern population and the highest coverage sample of the ancient sites, allowing for 0 to 3 migration edges.



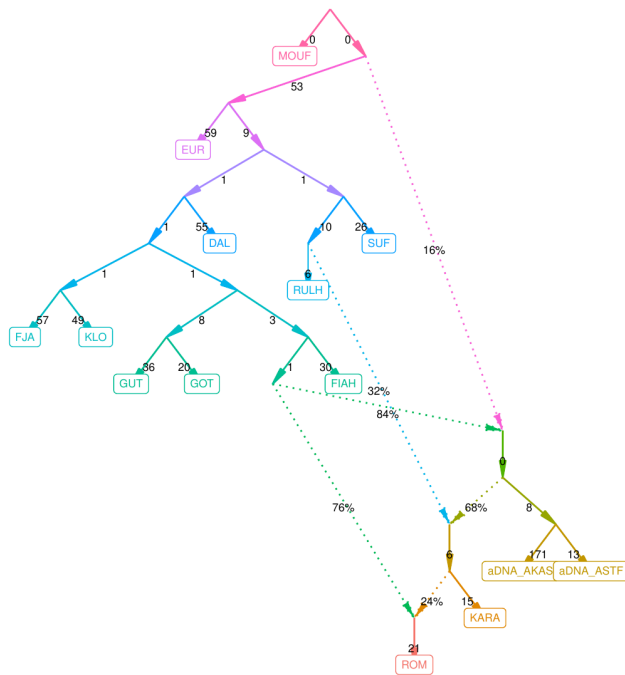
Supplementary Figure 10a: Best fitting admixture graph for $m=0$ and without constraints.



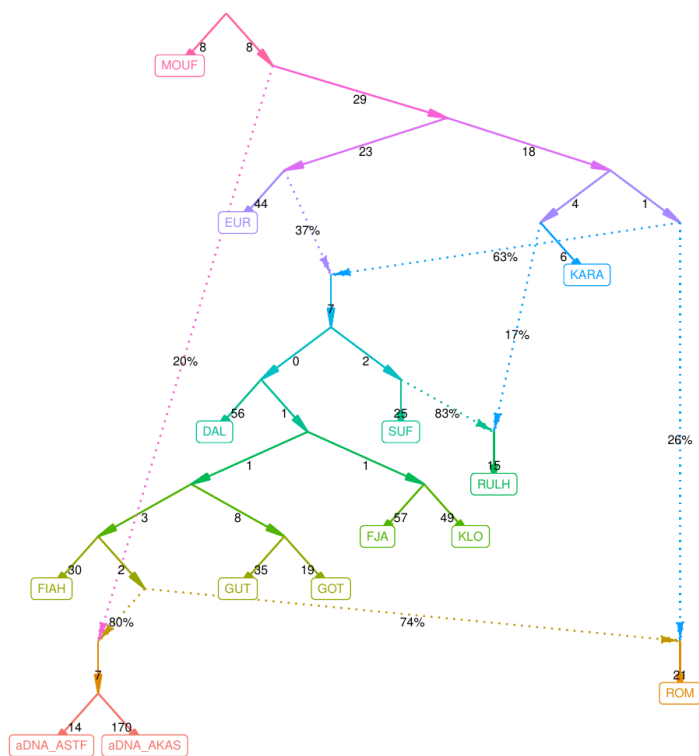
Supplementary Figure 10b: Best fitting admixture graph for $m=1$ and without constraints.



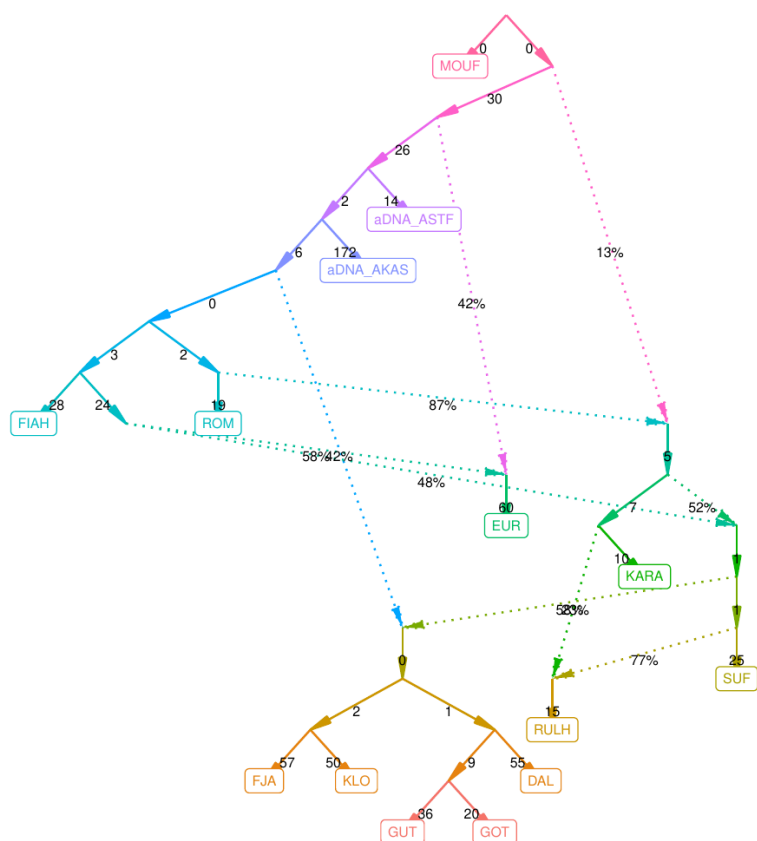
Supplementary Figure 10c: Best fitting admixture graph for $m=2$ and without constraints.



Supplementary Figure 10d: Best fitting admixture graph for $m=3$ and without constraints.



Supplementary Figure 10e: Best fitting admixture graph for $m=4$ and without constraints.

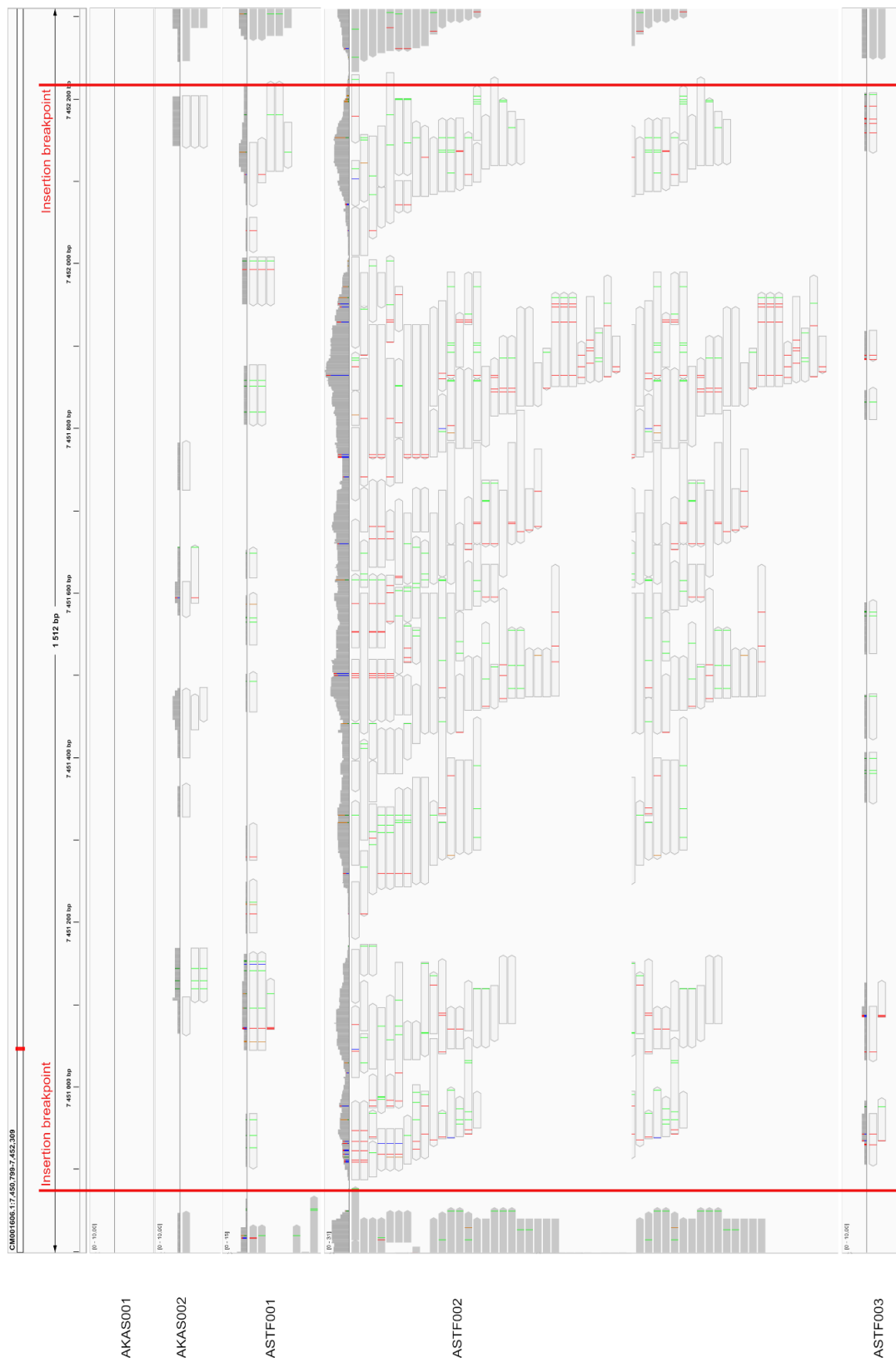


Supplementary Figure 10f: Best fitting admixture graph for $m=5$ and without constraints.



Supplementary Figure 11a: IGV (Robinson et al. 2011) plot of reads mapping to the ancestral ('hairy') reference genome version without the antisense *EIF2S2* retrogene into the

3' untranslated region (UTR) of the gene *IRF2BP2*. The insertion point is marked by a red line.



Supplementary Figure 11b: IGV plot (Robinson et al. 2011) of reads mapping to the derived ('wooly') reference genome version without the antisense *EIF2S2* retrogene into the 3' untranslated region (UTR) of the gene *IRF2BP2*. The insertion break points are marked by

Additional References in the Supplementary Information

- Davenport KM, Bickhart DM, Worley K, Murali SC, Salavati M, Clark EL, Cockett NE, Heaton MP, Smith TPL, Murdoch BM, et al. 2022. An improved ovine reference genome assembly to facilitate in-depth functional annotation of the sheep genome. *GigaScience* 11:giab096.
- Lagler DK, Hannemann E, Eck K, Klawatsch J, Seichter D, Russ I, Mendel C, Lühken G, Krebs S, Blum H, et al. 2022. Fine-mapping and identification of candidate causal genes for tail length in the Merinolandschaf breed. *Commun Biol* 5:1–13.
- Robinson JT, Thorvaldsdóttir H, Winckler W, Guttman M, Lander ES, Getz G, Mesirov JP. 2011. Integrative genomics viewer. *Nature biotechnology* 29:24–26.